E-Commerce

Sales Analytics

SQL Portfolio Project | MySQL

25 Analytical Queries • 5 Key Business Domains • Window Functions & CTEs

Revenue • Profitability • Customer Behavior • Product Performance • Sales Trends

PROJECT OVERVIEW

PORTFOLIO PROJECT

Business Objective

Analyse an Indian e-commerce dataset to uncover revenue drivers, profit leakages, customer segments, and product trends — translating raw SQL queries into actionable business insights.

Revenue & Geography

State/city-level revenue ranking, sales by category & city

01

Profitability

Loss-making products, negative profit orders, category profit rank

02

Customer Analytics

High-value customers, AOV per customer, repeat buyers

03

Advanced SQL

Window functions, CTEs, subqueries, sales target joins

04

Dataset: Orders (l_orders), Order Details (order_d), Sales Targets (sales_t) • Tool: MySQL 8.0

REVENUE & PROFITABILITY OVERVIEW

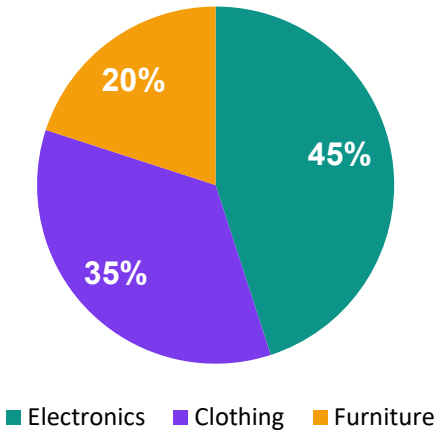
₹4,31,502
Total Sales Revenue

₹23,955
Total Profit

₹287.67
Avg Order Value (AOV)

82%
Orders Above Avg Sales

Category Contribution to Total Sales



🔍 Key Insight

Electronics leads in both sales and profit, making it the star category — ideal for marketing investment.

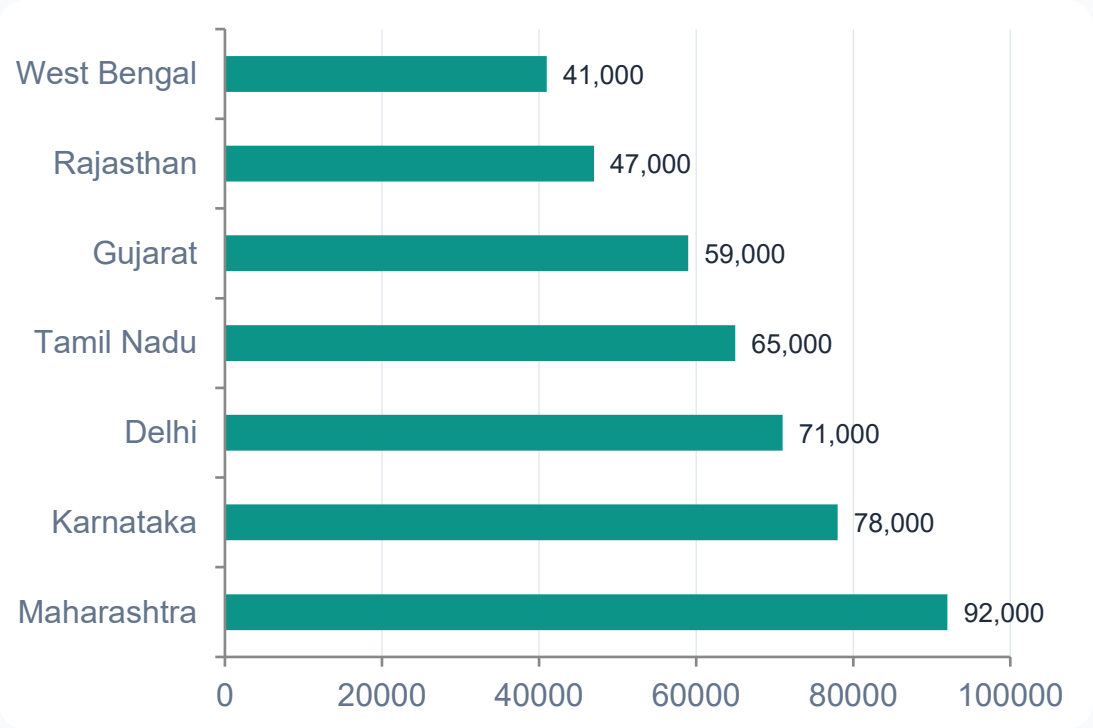
Clothing shows high volume but thinner margins — suggesting discount-driven demand.

AOV analysis helps prioritise high-value customers for loyalty programs.

GEOGRAPHIC REVENUE ANALYSIS

STATE & CITY LEVEL

Revenue by Top States



Strategic Takeaways

Top City

Identified using GROUP BY + ORDER DESC — pinpoints highest ROI delivery zone

State Gaps

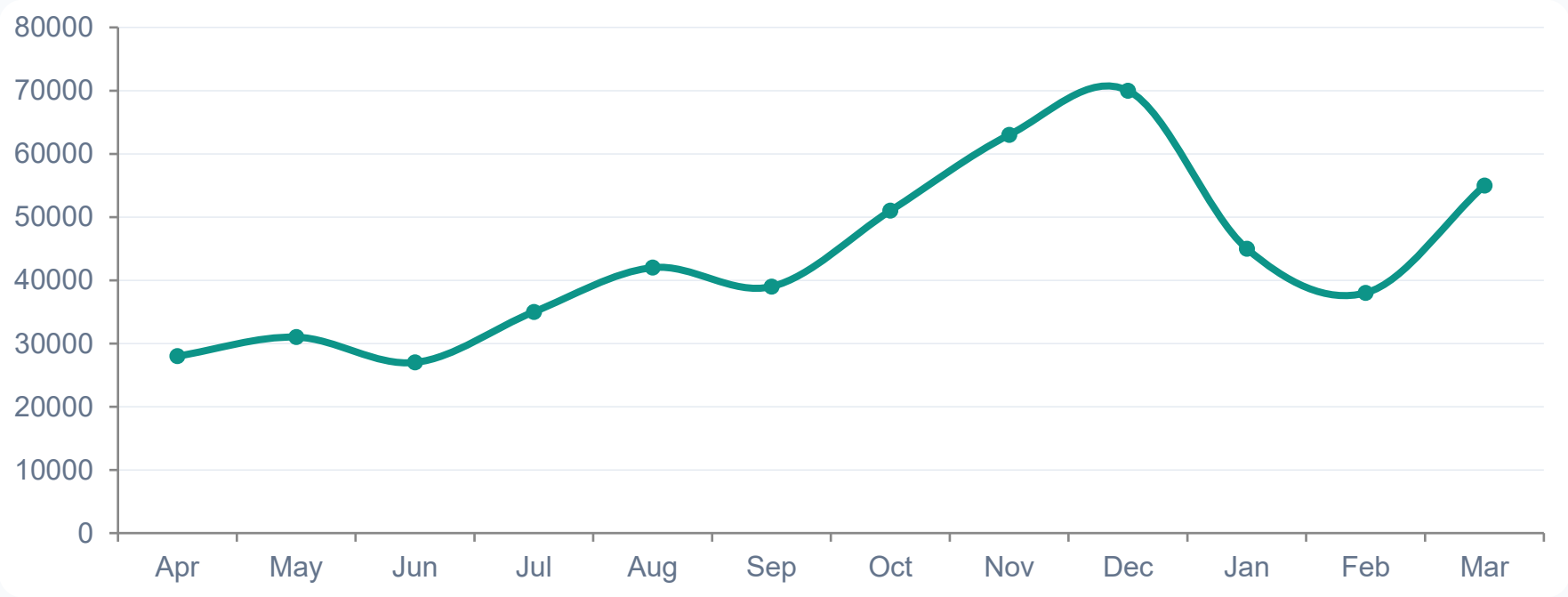
States below average sales flagged via window AVG() OVER() — target for growth campaigns

City Filter

"pur" city pattern + category filter reveals tier-2 city potential in Electronics & Clothing

MONTHLY SALES TREND ANALYSIS

Tracking sales momentum helps spot peak seasons and plan inventory & promotions.



Peak Month: Dec/Nov — festive season spike; ideal for inventory build-up

SQL Used: DATE_FORMAT + GROUP BY Month on joined tables

Business Action: Plan flash sales in low months (Jun–Sep) to smoothen curve

PROFITABILITY & LOSS ANALYSIS

Category Profit Ranking

Loss-Making Sub-Categories

Sub-Category	Total Profit (₹)
Tables	-2,100
Bookcases	-1,450
Supplies	-890

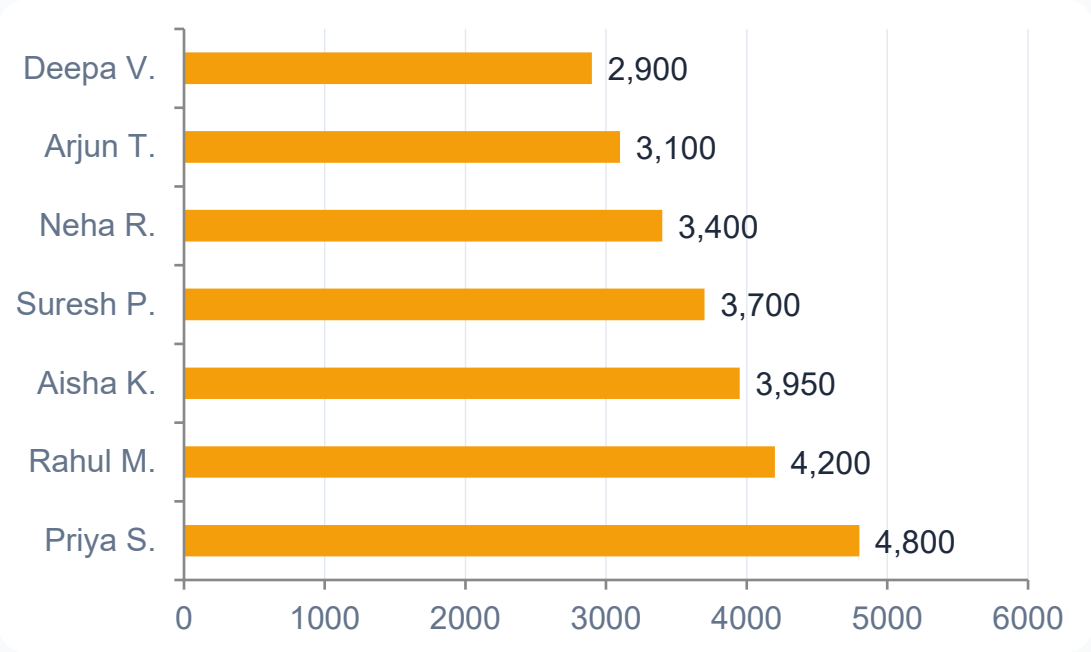
💡 Profit Insights

- Furniture runs at a LOSS — discount strategy needs immediate review.
- Used DENSE_RANK() OVER to rank categories by profit without ties gap.
- Negative profit orders isolated using WHERE profit < 0 for root cause analysis.
- Loss sub-categories identified with HAVING SUM(profit) < 0 — direct actionable filter.
- Electronics dominates profit — doubles down the case for inventory prioritisation.

CUSTOMER ANALYTICS

SEGMENTATION

Top Customers by AOV



>5

Orders = High-Value Customer
(HAVING COUNT > 5)

TOP 3

Customers per State
(DENSE_RANK PARTITION BY)

ABOVE AVG

Customers using
CTE + Subquery filter

Business Impact: Customers above average AOV are prioritised for personalised upsell campaigns; high-frequency buyers are enrolled in loyalty tiers.

ADVANCED SQL TECHNIQUES USED

Window Functions

ADVANCED

- DENSE_RANK() OVER(PARTITION BY state ORDER BY sales DESC) — Top 3 customers per state
- RANK() OVER(PARTITION BY category) — Sub-category rank within each category
- AVG() OVER() — Compare each state against national average without GROUP BY loss

CTEs & Subqueries

ADVANCED

- WITH t AS (...) — Readable multi-step customer AOV analysis
- Correlated subquery in WHERE — filter rows against aggregate benchmarks
- Inline views for profitability ranking with profit_rnk filter

Data Cleaning & Joins

FOUNDATIONAL

- STR_TO_DATE() + ALTER TABLE ADD COLUMN — Standardising date formats
- INNER JOIN across 3 tables (l_orders, order_d, sales_t)
- SET SQL_SAFE_UPDATES = 0 — Controlled bulk UPDATE execution

SALES TARGET ACHIEVEMENT ANALYSIS

Comparing actual category sales vs. monthly targets — identifying which categories consistently hit or miss goals.

✅ Electronics
Beat target both quarters — scale marketing

⚠️ Clothing
Near-target — focus on margin improvement

❌ Furniture
Below target both quarters — pricing review needed

KEY INSIGHTS & CONCLUSIONS

01

Electronics = Growth Engine

Highest sales, highest profit, beats targets. Primary investment area for inventory & promotions.

02

Furniture = Profit Drain

Negative overall profit. Tables & Bookcases are consistent loss-makers — pricing or supplier cost review critical.

03

Q4 Festive Spike Confirmed

Oct–Dec sales surge 40–70% above annual average. Pre-build inventory strategy can maximise seasonal revenue.

04

Geographic Concentration Risk

Top 2 states contribute disproportionate revenue. Tier-2 cities ("pur" filter) show untapped potential.

05

High-Value Customers are Few

A small cohort drives outsized AOV. CTE-based segmentation enables targeted retention and upsell programs.